Activity 5

The sides of a given square is 10 cm. The mid points of its sides are joined to form a new square. Again, the mid points of the sides of this new square are joined to form another square. This process is continued indefinitely. Find the sum of the areas and the sum of the perimeters of the squares formed through this process.

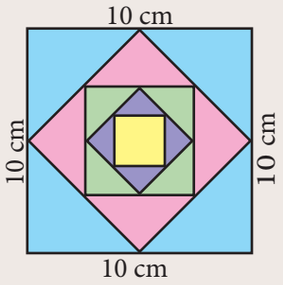


Fig.2.15

Example 2.51 Find the sum to n terms of the series $5 + 55 + 555 + \cdots$

Solution The series is neither Arithmetic nor Geometric series. So it can be split into two series and then find the sum.

$$5 + 55 + 555 + \cdots + n \text{ terms} = 5\left[1 + 11 + 111 + \cdots + n \text{ terms}\right]$$

$$= \frac{5}{9}\left[9 + 99 + 999 + \cdots + n \text{ terms}\right]$$

$$= \frac{5}{9}\left[(10-1) + (100-1) + (1000-1) + \cdots + n \text{ terms}\right]$$

$$= \frac{5}{9}\left[(10 + 100 + 1000 + \cdots + n \text{ terms}) - n\right]$$

$$= \frac{5}{9}\left[\frac{10(10^n - 1)}{(10-1)} - n\right] = \frac{50(10^n - 1)}{81} - \frac{5n}{9}$$

Progress Check

1. Is the series
   $3 + 33 + 333 + \ldots$
   a Geometric series?
2. The value of r, such that
   $1 + r + r^2 + r^3 \ldots = \frac{3}{4}$ is___.

Example 2.52 Find the least positive integer n such that $1 + 6 + 6^2 + \cdots + 6^n > 5000$

Solution We have to find the least number of terms for which the sum must be greater than 5000.

That is, to find the least value of n. such that $S_n > 5000$

We have, $S_n = \frac{a(r^n - 1)}{r - 1} = \frac{1(6^n - 1)}{6 - 1} = \frac{6^n - 1}{5}$

$$S_n > 5000 \Rightarrow \frac{6^n - 1}{5} > 5000$$

$$6^n - 1 > 25000 \Rightarrow 6^n > 25001$$

$\because$ $6^5 = 7776$ and $6^6 = 46656$

The least positive value of n is 6 such that $1 + 6 + 6^2 + \cdots + 6^n > 5000$.

Example 2.53 A person saved money every year, half as much as he could in the previous year. If he had totally saved ₹ 7875 in 6 years then how much did he save in the first year?

Solution Total amount saved in 6 years is $S_6 = 7875$

Since he saved half as much money as every year he saved in the previous year,

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